Actuators in Robotics & Automation

Focus on Electric Actuators

PC: https://www.robotshop.com/; Google Search Engine;

Actuators

- An Actuator is a device that converts energy into motion
- An actuator requires a **control signal** and a **source of energy**

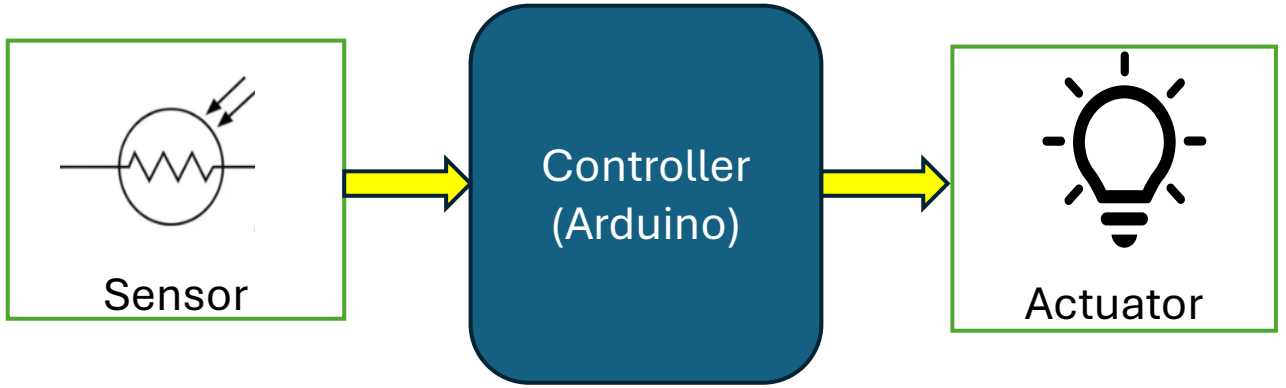
- **Energy sources:**
  - Electrical
  - Hydraulic
  - Pneumatic
  - Others (Thermal, Magnetic, shape- memory materials)



sensors = senses
controller = brain
actuator = **muscle**



Role of Actuators in Automation and Robotics



Sensors provide information

Controller makes decisions

Actuators carry out actions

Actuators in Automation and Robotics







Actuators drive the wheels for navigation and powers the arm for grasping and manipulation tasks.

Actuator Classifications

- **By energy source:**
  - Electric Actuators: Use electricity to generate motion (e.g. DC motor)
  - Hydraulic actuators: use pressurized liquid for very high force. (construction excavator, boom –lift)
  - Pneumatic actuators: use compressed air. (suction cups, conveyor gate, lift tables)
  - Other: Mechanical, Magnetics, Thermal…
- **By motion:**
  - Linear: Creates straight-line motion
  - Rotary: create rotational motion
- **By Control Method:**
  - On–off: simple start or stop
  - Proportional: allows intermediate levels
  - Continuous: precise adjustment across a wide range







Common Electric Actuators

- DC/AC Motors
- Stepper Motors
- Servo Motors
- Linear Electric Actuators

DC Motor

- Simple, low cost, rotary actuators
- **Applications:** Fan and blowers, portable tools, automobile,

Types:
Brushed DC Motor
Brushless DC Motor (BLDC)





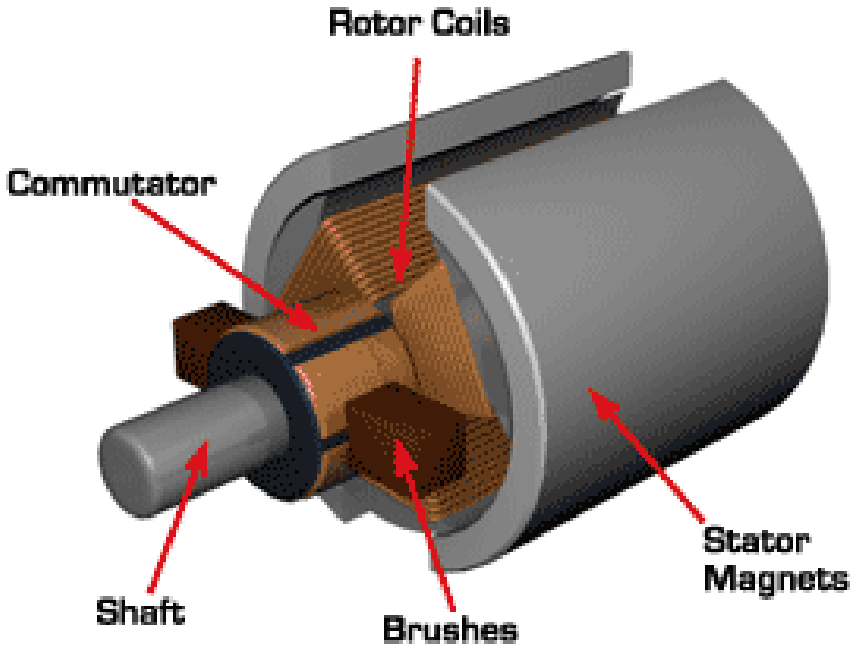






DC Motor Cross-sectional View

- **Basic Components**:
  - **Stator**: Fixed magnets create a constant magnetic field.
  - **Rotor (Armature)**: The rotating part inside the stator that carries the current-carrying coils.
  - **Commutator**: Split ring reverses current to keep rotor spinning.
  - **Brushes**: Supply electric current to the commutator.



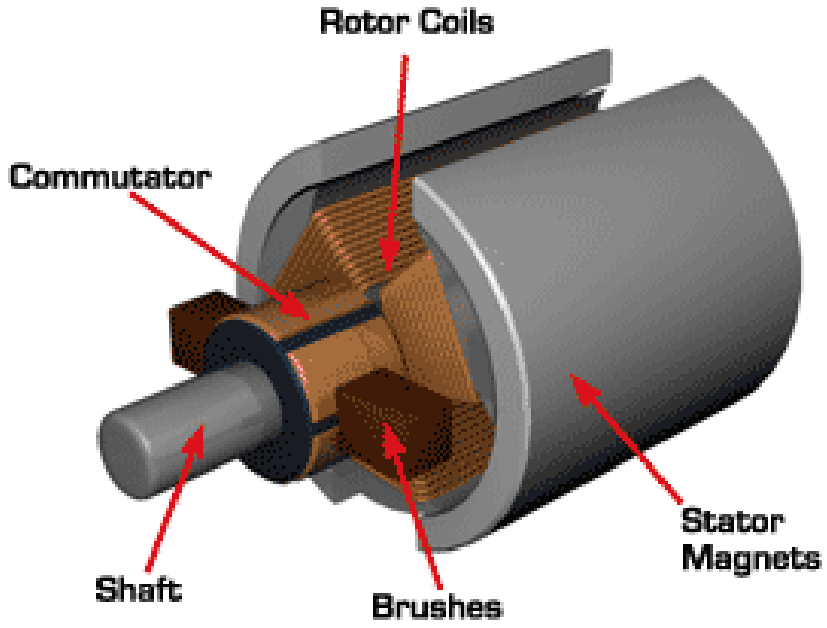
DC Motor Working Principle

Working Principle:

- DC voltage sends current through brushes to rotor's coil. When current flows in the coil, it produces its own magnetic field.
- Coil's magnetic field interacts with stator's field, creating force.
- This force causes the rotor to rotate.

Continuous Motion:

- Commutator switches current direction every half-turn.
- Ensures rotor's magnetic field aligns for ongoing rotation.

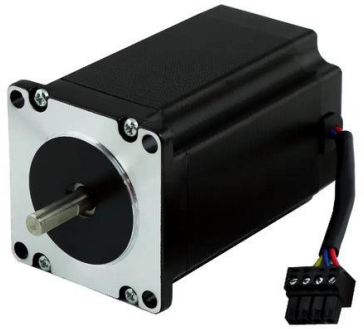


- Watch: Working Principle of DC Motor

Output:

- Spinning shaft powers mechanical systems (e.g., wheels, fans).
- Speed controlled by voltage; torque depends on current.

Stepper Motor

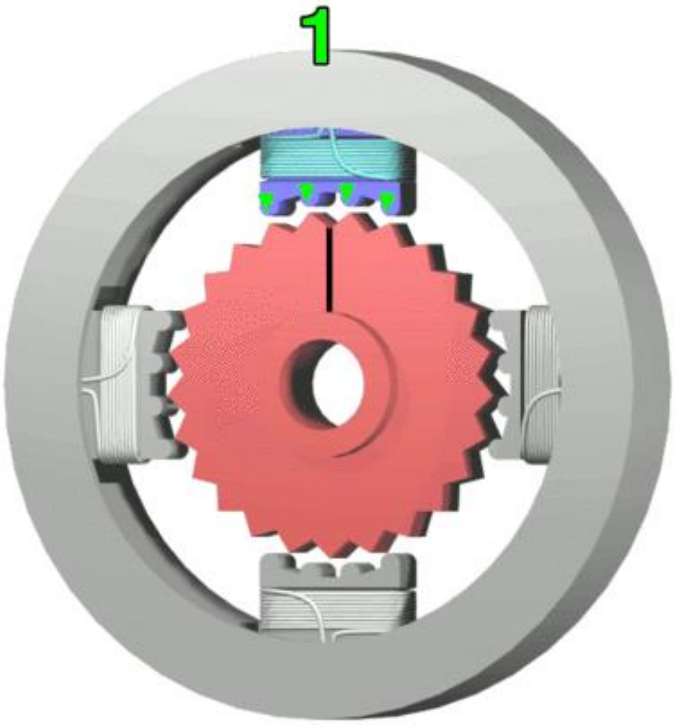


- Moves in **small**, **discrete steps**
- Open-loop positioning (no feedback)
- High precision, moderate speed, Cost-effective



- **Applications:** 3D printers, CNC, scanners

Stepper Motor



Components:

- **Rotor**: Permanent magnet or toothed iron core that rotates.
- **Stator**: Multiple coils (phases) act as electromagnets.
- **Driver Circuit**: Controls coil activation sequence.

Control and Precision:

- Pulses from a microcontroller dictate speed and direction.
- Micro stepping allows smaller, smoother steps by varying current.
- Holds position when coils are powered, no feedback needed.

Stepping Process:

When a coil in the stator is energized with electric current, it creates a magnetic field that attracts the rotor's poles, causing it to align with that coil.

By switching the current to the next coil in a specific order, the rotor moves a fixed angle (called a "step") to align with the new magnetic field.

Servo Motor



- DC motor + **gear + feedback sensor + controller**
- Controlled by signal (e.g., PWM pulse width)
- Provides precise angular position

- **Applications**:
  - Precise angular control for robotics, RC vehicles, automation.
  - High torque and stable positioning for small movements.



Linear Electric Actuators

- Rotary → Linear motion conversion

Motor Types: DC or Stepper or Servo motor that provides rotational power.



Ball Screw Linear Actuator System



Types:

1. **Guided-linear Rails: Lead Screw or Rack-and-pinion or belt-driven Systems**: Converts the motor's rotation into linear motion using threaded screw / rack-and-pinion or belt-driven systems. Suited for automation, sliding doors, or lighter loads.

2. **Rod Actuator /Telescopic**: The part that extends or retracts to move the load

3. **Solenoids:** Electromagnetic coil moves plunger for short, rapid strokes. Used in locks, valves, or automotive systems.



Linear Actuators Applications

Applications:

- **Industrial**: Moving heavy machinery, conveyor alignment, or gantry robots.
- **Consumer**: Adjustable desks, recliners, or home automation (e.g., window openers).
- **Automotive**: Seat adjustments, trunk lifts, or throttle control.
- **Medical**: Hospital beds, prosthetic devices, or surgical equipment.
- **Solenoids**: Specific uses like electromagnetic locks, fuel injectors, or camera shutters.

Electric Actuators

- Why electric actuators are most widely used in robotics?

Key Benefits:

- Precise & easy to control
- Compact & widely available
- Integrate with electronics
- Low maintenance

Limitations:

- Lower torque/force,
- Overheating at high loads

Wrap-Up

- **Actuators** = muscles of automation
- **Electric actuators** = precision, easy integration
- **Key difference**: DC (continuous), Stepper (steps), Servo (feedback, position)

Quick Question:

Which actuator would you choose for ?

Robotic arm joint?

3D printer head?

Robot Wheel ?

Actuators in Robotics

- **Wheels:** DC / BLDC
- **Arms/Joints:** Servos, BLDC
- **Grippers:** Servo / Linear actuators
- **Drones:** BLDC for propellers
- **3D printer/Gantry Robot:** Stepper Motor



https://www.alibaba.com/